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1. Introduction — The Laser Gyro Revolution

The laser gyro has caused a **technological revolution** in inertial reference and navigation systems. This solid-state, high-precision angular rate sensor is ideally suited for a highly reliable **strap-down configuration** — it eliminates gimbals, bearings, torque motors, and other moving parts.

Inertial Navigation means determination of a vehicle's location without external references. **Strap-down inertial navigation** goes further — navigation without a mechanically stabilised platform. The laser gyro/rate sensor and high-speed microprocessors allow a *mathematical* (rather than mechanical) stable platform.

2. Inertial Reference Unit (IRU) — Outputs

The IRU is the heart of the IRS. It provides all required inertial reference outputs for the aircraft's avionics. Primary sources of information: **three laser gyros** and **three inertial accelerometers** — plus initial position, barometric altitude, and TAS entered externally.

Category	Output Parameters
Primary Attitude	Pitch and Roll
Heading	True and Magnetic
Accelerations	Lateral, Longitudinal, Normal
Angular Rates	Pitch, Roll, Yaw
Inertial Velocity	N/S, E/W, Ground Speed, True Airspeed, Vertical Rate
Position	Latitude, Longitude, Inertial Altitude
Wind Data	Wind Speed, Wind Angle, Drift Angle
Calculated Data	Flight path angle/acceleration, along/across track accel, inertial pitch/roll rate, vertical accel, potential vertical speed

Why barometric altitude and TAS are needed:

- **Baro altitude** — stabilises the vertical navigation channel; prevents vertical velocity and inertial altitude from drifting
- **TAS** — allows the IRU to calculate **wind speed and direction** (wind = inertial velocity − TAS vector)
- **Initial position** — present position is calculated from distance and direction travelled from start position; the IRU cannot determine position from scratch

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IRS inertial information is used by: Flight Management Computer, Flight Control Computer, Thrust Management Computer, Stability Augmentation System, Weather Radar, Anti-skid / Auto-brake, ADI, HSI, VSI, RDMI, FDR.

3. The Laser Gyro — Principle (Sagnac Effect)

The laser gyro measures rotation by comparing two laser beams created and directed to travel in **opposite directions** within a very narrow triangular tunnel.

The Sagnac Effect:

- When stationary: both beams travel equal path lengths → cancel each other out → no output
- When rotating: one beam takes longer (longer path) and the other takes shorter → the path lengths change → **frequency changes**
- The frequency difference between the two beams = measure of **angular rate**
- This frequency difference is easily and accurately measured along optical paths

> *"The change in frequency, caused by the change in path length due to rotation of the gyro, is known as the **SAGNAC effect**."*

Lasing is achieved by running high voltage through **helium–neon gas** between anodes and cathode, transforming atoms into light in the **pinkish-orange** part of the visible spectrum.

4. Construction and Operation

Construction

- **Triangular block** of temperature-stable glass
- Small tunnels drilled parallel to the perimeter of the triangle
- **Three mirrors** in each corner to achieve rotational path for two counter-rotating beams

The Three Mirrors — Specific Roles

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Mirror	Function
Mirror 1	Makes micro-adjustments to keep the physical light path accurately aligned
Mirror 2	Partially transparent — allows laser light to be detected on photo-cell detectors. Includes a prism that redirects/flips the beam to cause interference with the direct beam.
Mirror 3	Standard high-reflectivity mirror

Detection — Fringe Pattern

1. Mirror 2's prism redirects beam → both beams meet → **interference** (alternately cancel and reinforce)
2. This creates a **fringe pattern**
3. A **photoelectric cell** detects the direction and speed of fringe pattern movement
4. Direction of pattern movement = direction of rotation
5. Speed of pattern movement = rate of rotation → converted to angular rate signal

Path length and frequency: If path length decreases, light is compressed → frequency *increases*. If path length increases, frequency *decreases*.

5. Limitations — Drift, Accuracy, Lock-In, Dither

Drift

Principal error source, as with conventional gyro INS. In the laser gyro, noise is the cause — derived almost entirely from **imperfections in the mirrors and coatings** (not mechanical bearing imperfections as in a conventional gyro).

Accuracy

Accuracy is directly influenced by the **length of the optical path**. A small percentage increase in path length leads to a **substantial increase in accuracy**.

Lock-In (Laser Lock)

Problem: At very low rotation rates, **back-scattering** between the two beams causes them to synchronise → output frequency drops to zero → rotation is not detected → undesirable errors introduced.

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Dither Motor (Piezo-electric)

Solution — Dither:

- A **piezo-electric dither motor** vibrates the laser ring about its input axis through the lock-in region
- This "breaks" the lock-in by ensuring the gyro never stays at zero rotation rate
- The dither motor motions are **decoupled from the output** of the ring laser gyro — so they do not contribute error to the navigation solution
- The photoelectric sensor can then detect smaller fringe pattern movements

DGCA exam key fact: Dither is used to **break the frequency lock** (lock-in) which would prevent small rotational rates from being sensed. It does NOT enhance accuracy at all rates or increase maximum sensing rate.

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6. Platform vs. Strap-Down (IRS)

```
graph TD
  A["INS — Platform System"] --> B["3 rate-integrating gyros
3 accelerometers
on mechanically stabilised gimbal platform"]
  C["IRS — Strap-Down"] --> D["3 laser gyro/rate sensors
3 accelerometers
bolted directly to aircraft chassis"]
  D --> E["High-speed microprocessor
maintains stable platform
MATHEMATICALLY"]
```

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Aspect	INS (Platform)	IRS (Strap-Down)
Gyro type	Rate-integrating (mechanical)	Ring laser gyro (optical)
Platform	Mechanical gimbals, motors, bearings	Mathematical (no moving parts)
Gravity compensation	Mechanical levelling of platform	Microprocessor subtracts local gravity from vertical acceleration
Earth rate compensation	15.04°/hr torquing of gyros	Same — compensated at 15.04°/hr
Transport rate / Schuler tuning	V/R torquing; 84.4-min Schuler cycle	Same — Schuler tuning required for oscillation errors
Calibration	Manual	Automatic (computer-based)
Reliability	Lower (many moving parts)	Much higher (solid-state)

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7. Alignment — Establishing the Trihedron

THE AIRCRAFT MUST NOT BE MOVED DURING ALIGNMENT.

Alignment sequence:

1. **Finding True North:** Aircraft stationary → the only rate of change is Earth rotation → system detects Earth rotation vectors → True North computed
2. **Latitude verification:** Operator enters initial latitude. Computer assesses rotational vectors it is experiencing and compares with entered latitude. If there is a discrepancy, crew is alerted.
3. **Memory function:** The IRS **remembers its position at landing** — on startup it will indicate any errors in initial position input (lat or long) to the crew
4. **Mathematical levelling:** Computer completes full mathematical levelling process
5. Process = **"Establishing the Trihedron"**

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Unlike the INS, the IRS (with memory) can flag discrepancies between the entered position and its remembered landing position. This is an additional safeguard against gross position entry errors.

8. Advantages of IRS over INS

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Advantage	Detail
Activation time	Almost no spin-up time — **~1 second activation** for the rate sensor
Manoeuvring	Insensitive to 'g', attitude, rolling and pitching manoeuvres
Construction	Mechanically simple, highly reliable (no gimbals, bearings, torque motors)
Dynamic range	Wide dynamic range
Drift	Very small drift rates — **greatest errors induced by the operator** (position entry)

9. INS vs. IRS — Key Comparison Summary

Both systems share:

- Double integration principle (acceleration → velocity → distance)
- Initial position must be entered manually
- Aircraft must be stationary during alignment
- Schuler tuning (84.4-min period)
- Earth rate compensation at 15.04°/hr
- Transport rate compensation (V/R)
- Coriolis and centrifugal compensation
- Accumulated position error with time

IRS advantages over INS:

- No mechanical platform → more reliable, lighter, smaller
- Faster alignment (~10 min vs. INS ~15 min)
- Laser gyros have very small drift rates
- Memory function for position verification on restart
- Insensitive to vibration and manoeuvres

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Practice Question & Detailed Answer

Q1. Dither is used in a laser gyro in order to:

a. Enhance the accuracy of the gyro at all rotational rates

b. Increase the maximum rotational rate that can be sensed by the gyro

c. Stabilize the laser frequencies at peak power output

d. Break the frequency lock which would prevent small rotational rates from being sensed by the gyro

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✅ Correct Answer: **D**

At very low rotation rates, back-scattering between the two counter-rotating laser beams causes them to "lock" together (synchronise), dropping the output frequency to zero — so the gyro cannot sense those very small rotations. The piezo-electric dither motor vibrates the laser ring about its input axis, taking the gyro through the lock-in region and preventing synchronisation. The dither motion is then electronically decoupled from the output so it does not introduce navigation error.

Why the others are wrong:

- **A:** Dither is specifically for breaking lock-in at *low* rates, not for enhancing accuracy at all rates.
- **B:** Maximum rotation rate sensing is a function of the optical path length and detector bandwidth — not dither.
- **C:** Laser frequency is stabilised by path length adjustment (mirror 1 micro-adjustment), not dither.

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This is the single most-examined point from Chapter 19. Remember: Dither = breaks lock-in = allows sensing of small rotational rates. The word "dither" itself implies vibrating/oscillating to prevent sticking.